PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Project Information (Cybersecurity Professional)	**Kirtana Minoy**
Program / Batch	B084-Networkwalks
Date	11 November 2026
Modules Completed	W2-PM1 – (Multiple Kali Tools) W2-PM5 – (Zenmap Scanning)
Client / Target	1.Networkwalks (secured written permission already) 2.My own LAN network
Permission Secured from Client?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3: In progress

1.Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks. All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in the report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner ,dates ,name servers).
WhatWeb	Fingerprint web technologies (server, CMS, plugins, IP)
NSLookup	Resolve the domain name to its IP address using DNS
cURL	Retrieve and inspect HTTP response headers.
WAFW00F	Detects the HTTP response headers of the website.
DNSRecon	Enumerate all DNS records (SOA, NS, A, AAAA, MX, TXT, and SRV).
Zenmap / Nmap 7.99	Scan the local subnet to find live hosts , Ips and MAC addresses
Windows CMD	Local IP and MAC address Identification.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target.

First, I used WHOIS to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used WhatWeb to identify technologies used by the website. The results identified WordPress 7.0.4 and WP Download Manager 3.3.58, along with other information exposed by the website.

Using Nslookup, I resolved the domain name to its IP address. The provided result identified 192.232.216.135.

I used Curl with the -I option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint /wp-json/.

Next, I used Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs).

Finally, I used DNSRecon to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap to perform network discovery on my local VirtualBox network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the ip -br addr command in Kali Linux to identify my network interface and local IP address. My active interface was eth0, with the IP address 10.0.0.3/24. I then entered the subnet 10.0.0.0/24 into Zenmap and selected the Ping Scan option to identify active hosts.

The scan was performed on 10 November 2026 using Nmap 7.99. It scanned 256 IP addresses in approximately 3.08 seconds and identified three live hosts:

- 10.0.0.1
- 10.0.0.2
- 10.0.0.3

Two hosts returned MAC addresses:

- 10.0.0.1: MAC address 52:55:0A:00:00:01 (vendor identified as Unknown).

- 10.0.0.2: MAC address 08:00:27:AE:38:62 (Oracle VirtualBox virtual NIC).
- 10.0.0.3: This was the Kali Linux machine used to perform the scan.

After completing the scan, I opened the Topology section in Zenmap, where the discovered hosts were displayed in a graphical network diagram using circles and connecting lines. This provided a visual representation of the network discovery results.

Note: My scan was a Ping Scan (host discovery), so it identified active hosts but did not determine open ports or running services.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1.4 and WordPress Download Manager 3.3.76.	Attackers may use exposed technology and version information to identify software requiring further security review.	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135.	Provides information about the network location of the web service.	● Low
3	HTTP technical information exposed	cURL returned HTTP response headers and exposed the /wp-json/ endpoint.	May assist technology fingerprinting and further enumeration.	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs) WAF.	Reveals information about the web application's security architecture.	● Low
5	DNS infrastructure information exposed	DNSRecon identified 16 DNS records, including SOA, NS, MX, TXT and SRV records.	DNS information can help build a broader infrastructure profile.	● Medium
6	Multiple live hosts visible on local network	Zenmap identified three live hosts: 10.0.0.1, 10.0.0.2 and 10.0.0.3 in the 10.0.0.0/24 network.	Unknown or unauthorised devices may potentially be present on a network.	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules. Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. Review publicly exposed technology information
Organizations should regularly review what information about their web technologies, CMS platforms and plugins is publicly visible.

2. Keep software updated
WordPress, WordPress Download Manager, Bootstrap, jQuery and other web technologies should be regularly updated and reviewed against current security advisories.

3. Review HTTP headers
HTTP response headers should be reviewed to identify whether unnecessary technical information is being exposed through server responses.

4. Review DNS records regularly
DNS records should be checked periodically to ensure that only required domain, mail and service-related information is publicly available.

5. Properly configure and monitor the WAF
The detected ModSecurity (SpiderLabs) WAF should be properly configured, regularly updated and monitored to help protect the website against common web-based attacks.

6. Perform regular internal network discovery
Organizations should periodically scan their own networks to identify active devices and maintain visibility of connected systems.

7. Investigate unknown devices
Any unexpected or unidentified device discovered during network scanning, such as the host identified at 10.0.0.1, should be investigated and verified.

8. Maintain network documentation
Network topology, IP addresses and device information should be documented and updated regularly to support effective network monitoring.

9. Perform security testing with authorization
Reconnaissance and network scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

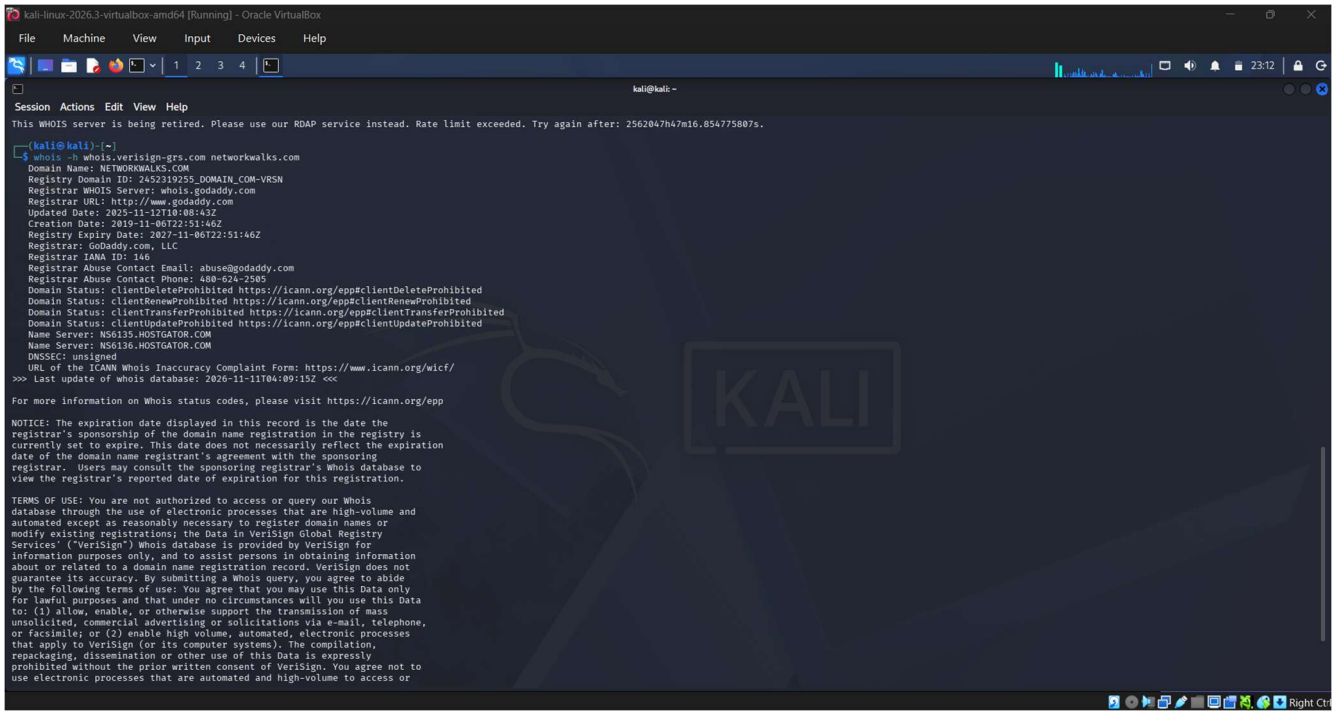
In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information. In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

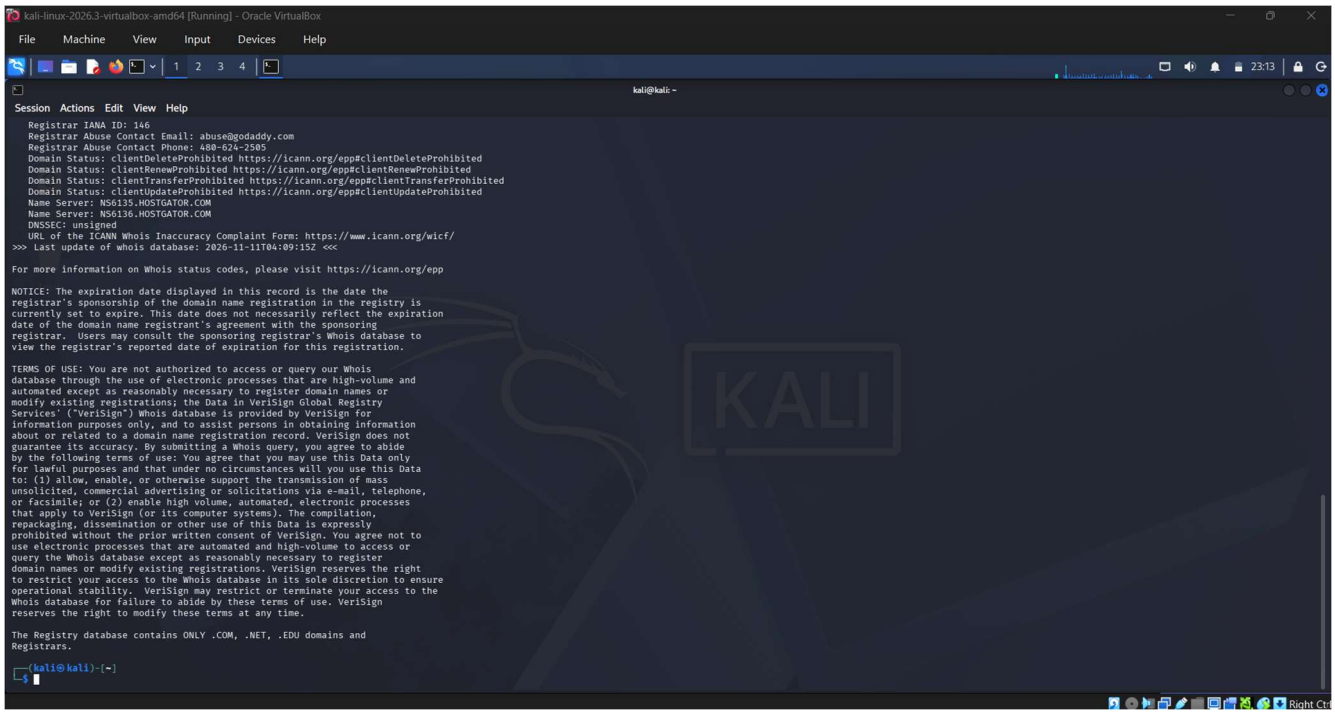
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected






```
(kali㉿kali)-[~]
$ whatweb https://networkwalks.com
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB8amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]

(kali㉿kali)-[~]
$
```

```
(kali㉿kali)-[~]
$ nslookup networkwalks.com
Server:         172.20.10.1
Address:        172.20.10.1#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
Name:   networkwalks.com
Address: 64:ff9b::c0e8:d887

(kali㉿kali)-[~]
$
```

```
(kali㉿kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=5018b2afcbe7bf797bb4b7b578382c61; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Wed, 11 Nov 2026 04:22:05 GMT
server: Apache
```

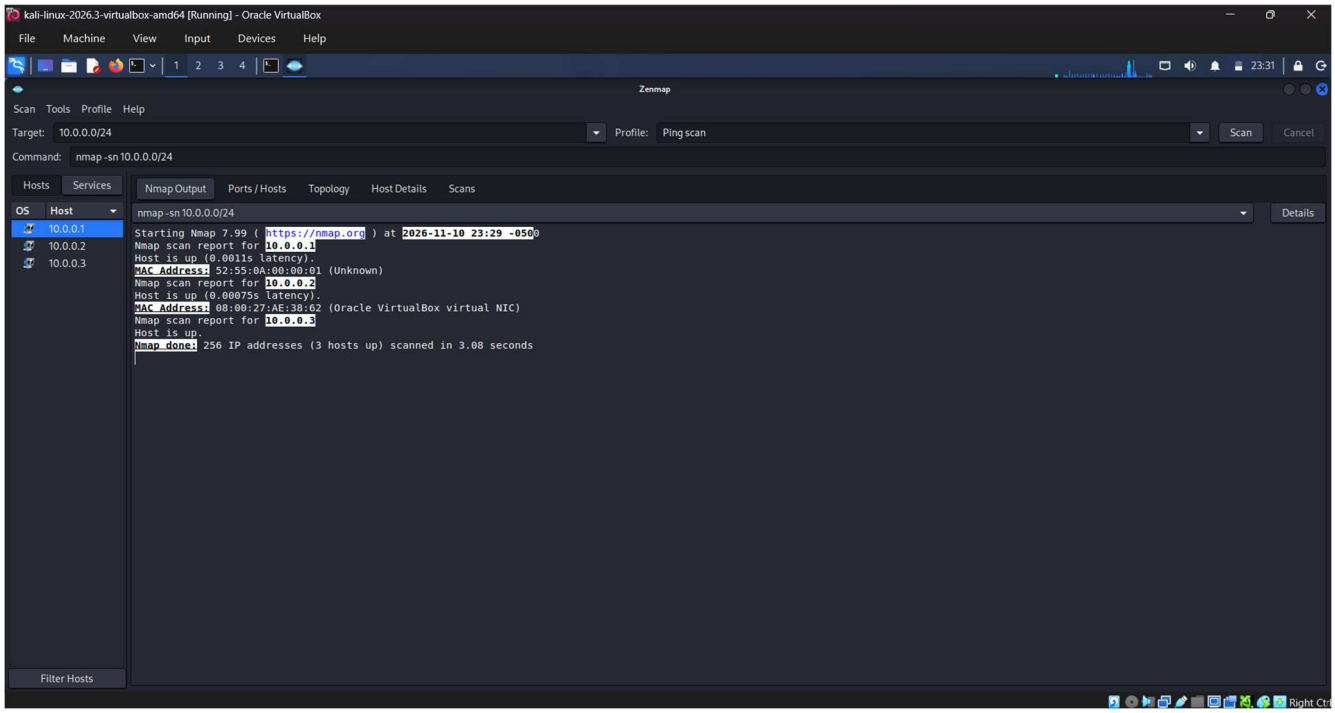
```
(kali㉿kali)-[~]
$ wafw00f https://networkwalks.com

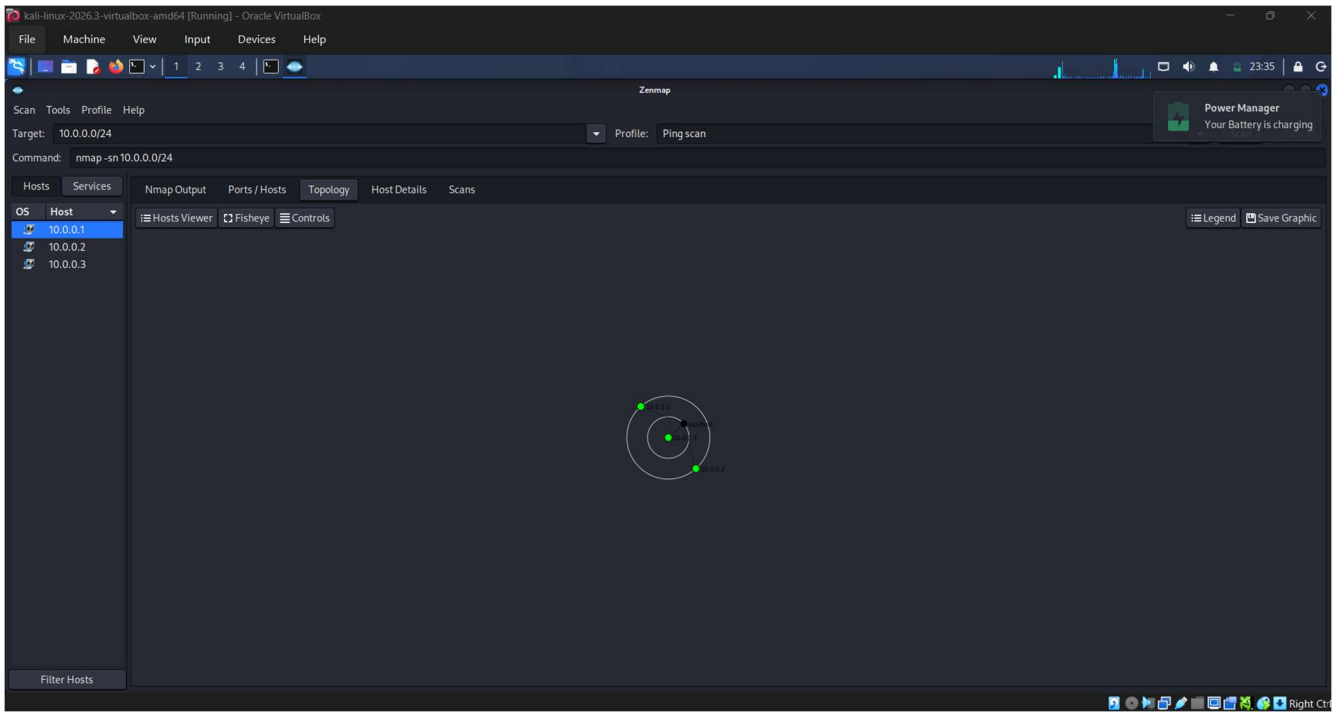
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/,___ /`                    |___|___|___|___|___|___|___|___|___|___|
\\   \\

                    ~ WAFW00F : v2.4.2 ~
        ~ Sniffing Web Application Firewalls since 2009 ~

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
```

```
┌──(kali㉿kali)-[~]
└─$ dnsrecon -d networkwalks.com
2026-11-10T23:25:29.569844-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-10T23:25:29.572812-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-10T23:25:31.392283-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-10T23:25:32.594348-0500 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-11-10T23:25:32.594615-0500 INFO     SOA ns6135.hostgator.com 64:ff9b::3257:9057
2026-11-10T23:25:34.348030-0500 INFO     NS ns6136.hostgator.com 192.232.216.131
2026-11-10T23:25:34.887638-0500 INFO     Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-10T23:25:34.887948-0500 INFO     NS ns6136.hostgator.com 64:ff9b::c0e8:d883
2026-11-10T23:25:34.890638-0500 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-10T23:25:35.522001-0500 INFO     Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-10T23:25:35.522270-0500 INFO     NS ns6135.hostgator.com 64:ff9b::3257:9057
2026-11-10T23:25:41.129612-0500 INFO     MX mail.networkwalks.com 192.232.216.135
2026-11-10T23:25:41.130613-0500 INFO     MX mail.networkwalks.com 64:ff9b::c0e8:d887
2026-11-10T23:25:41.169169-0500 INFO     A networkwalks.com 192.232.216.135
2026-11-10T23:25:41.169674-0500 INFO     AAAA networkwalks.com 64:ff9b::c0e8:d887
2026-11-10T23:25:46.751810-0500 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-10T23:25:46.753129-0500 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-10T23:25:48.328539-0500 INFO Enumerating SRV Records
2026-11-10T23:26:04.076264-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-10T23:26:04.076789-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-10T23:26:04.077415-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-10T23:26:04.078780-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-10T23:26:04.078978-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-10T23:26:04.079167-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-10T23:26:04.079314-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-10T23:26:04.079401-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-10T23:26:04.079570-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:ff9b::b85e:cb0e 443
2026-11-10T23:26:04.079712-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:ff9b::b85e:cc0f 443
2026-11-10T23:26:04.079791-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:ff9b::b85e:cc09 443
2026-11-10T23:26:04.079925-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:ff9b::b85e:cb0b 443
2026-11-10T23:26:04.080111-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:ff9b::b85e:cc0e 443
2026-11-10T23:26:04.080213-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:ff9b::b85e:cb0f 443
2026-11-10T23:26:04.080349-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:ff9b::b85e:cc0b 443
2026-11-10T23:26:04.080496-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:ff9b::b85e:cb09 443
2026-11-10T23:26:04.084764-0500 INFO 16 Records Found
2026-11-10T23:26:04.085323-0500 INFO Completed enumeration for domain: networkwalks.com
```





-End-

Project Information

Program Name : Cybersecurity program at Networkwalks | Week : 02 | Repository : Github

Github repository :

https://github.com/Kirtana24/Week-02-Footprinting-Reconnaissance-Network-Scanning